AT RISK



THE DELUGE BEGAN WITHOUT WARNING.

No rain, no bursting of a known glacial lake, not even an earthquake. High above the Nepal-China frontier on August 26, a slab of the mountain itself gave way. Satellite images indicate that the bedrock, and the glacier ice resting on it, detached at about 17,000 feet near the Langtang Himal and crashed to a valley floor roughly 4,000 feet below. The rupture scar stretched about 1.4 kilometres, measuring nearly 10 cricket stadia end to end. As that colossus hit the gorge, the impact shook the earth strongly enough to register as a magnitude-5.2 seismic event. Initially thought to be an earthquake, scientists later established that the impact of the fallen mass had generated the tremor.

Then came the black monster. Rock, shattered ice, water and mud fused into a fast-moving slurry that tore down the Lhende Khola river along the Nepal-Tibet border and into the Bhote Koshi-Trishuli river corridors at the Rasuwa border post. Chinese geologists estimated the debris flow at 180 kmph, or 50 metres a second. It scoured the narrow valleys, swallowed border posts, bridges, roads, power installations and settlements, and altered the river channel more than 100 km downstream. At Galchhi, 50 km north of Kathmandu, the river reportedly rose nine metres in half an hour. Survivors do not recount an ordinary flood; their memory is of a rising wall of liquid